From Leakage Events to QEC Syndromes

Simulation and source localization

Synthetic detector-level injection | Rotated surface code | Single-shot coordinate estimation

Introduction

Can QEC detection records reveal where a leakage event started?

We inject localized leakage-like disturbances into simulated detection records, then estimate the source qubit coordinates from the records alone.

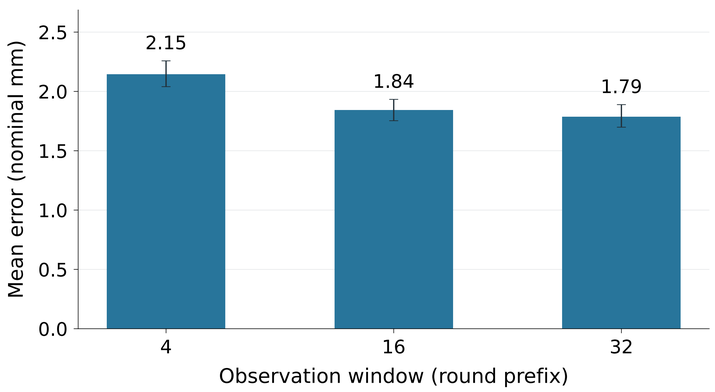
1 Leakage-to-syndrome pipeline

1. Sample a noisy surface-code circuit.
2. Choose a source qubit and an onset round.
3. Transport and decay leakage on the code graph.
4. Randomize affected detection records.
5. Estimate the source coordinates (x, y).

Input: responding checks across time.
Target: the source data-qubit coordinates.
True location, onset and strength are not inputs.

Detector-level injection, not a qutrit circuit model.
Transport = 0.123/round; decay = 7.63 rounds.

2 Longer observation



Longer observation reduces mean error.
The gain from 16 to 32 rounds is small.

Matched events and onset times; a separately trained SVR for each observation window.

Evaluation design

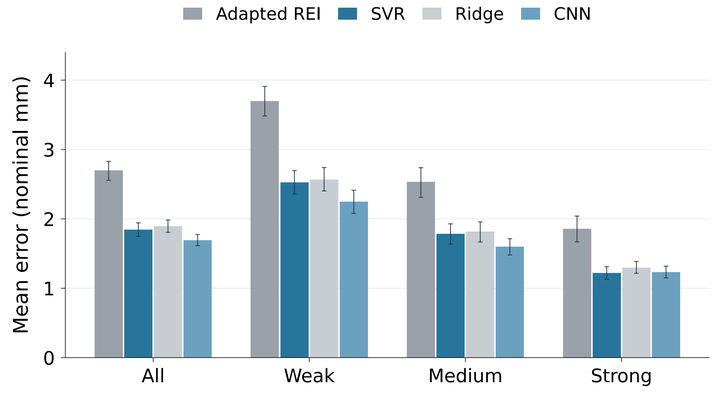
Primary: d=5, 16-round prefix (15 internal).
864 training + 180 validation events per replica.
540 test events; 3 replicas; 2 separate shots/event.
Error bars: 95% event-bootstrap intervals.

Main result

31.6% lower mean error

2.695 → 1.843 nominal mm

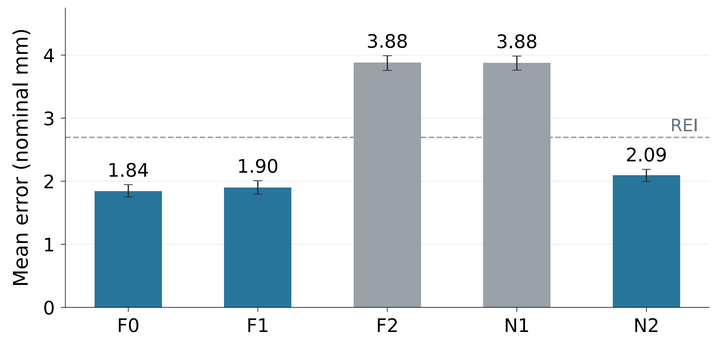
Adapted REI → SVR | Same test records



CNN has the lowest overall mean error (1.692).
SVR and CNN are close for strong events.

Weak / medium / strong: corruption probabilities
0.15 / 0.30 / 0.50, not measured leakage strengths.
REI answers 66.7%; other cases use a fixed fallback.

3 Which information matters?

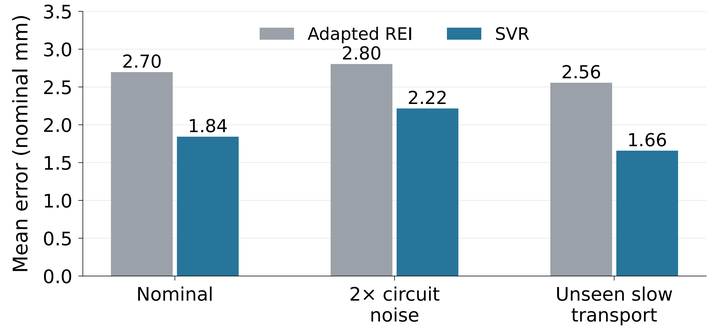


F0 Spatial + temporal F1 Spatial only
F2 Temporal only N1 Shuffled check positions
N2 Shuffled time order

Spatial information is the main localization cue.
Shuffling time order also increases error.

F1 is close to F0; their paired difference interval includes zero. Each feature variant is retrained.

4 Noise and propagation shift

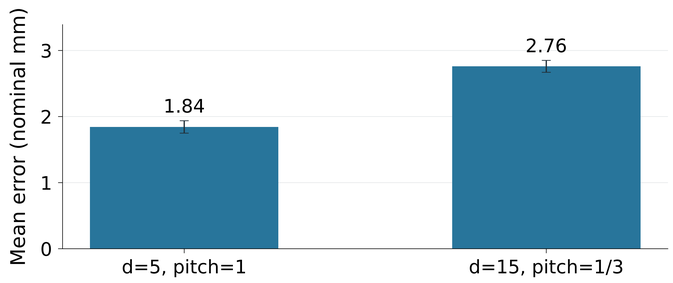


SVR remains more accurate than adapted REI in both tested shifts, without retraining.

Nominal models and quiet reference are frozen.
Unseen transport = 0.03/round (nominal: 0.123).

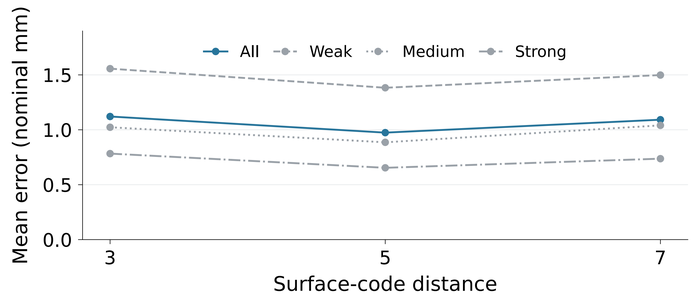
5 QEC layout as sensor geometry

A Higher density, fixed area



Higher density worsens SVR at fixed training size.
Same area and 25 source positions; more checks and background noise. Physical spread is not fixed.

B Wider coverage, fixed spacing



All: 1.121 → 0.974 → 1.092. Wider is not always better.
Common nine-source region; separate from the main test.

Scope: synthetic data-qubit sources; source sites shared across train/test. Nominal mm is an assumed scale. No measured-response or decoding validation. Gains are specific to the REI comparator.